ASTANA IT UNIVERSITY | HIGHER EDUCATION PEDAGOGY • MASTER'S DEGREE PROGRAM • DEPARTMENT OF GENERAL EDUCATION DISCIPLINES

GROUP: SSE-2601 | ACADEMIC YEAR 2026

CRITICAL ANALYSIS & PEDAGOGICAL INVESTIGATION

# THE ICEBERG OF LEARNING

## What Lies Beneath a High Grade?

PRIMARY RESEARCH QUESTION

**"Do high grades necessarily reflect actual understanding and the ability to apply knowledge?"**

*Focus: Evaluating the divergence between surface test metrics and multidimensional cognitive mastery.*

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### SECTION 1 • INTRODUCTION & BACKGROUND

Pedagogical Significance

INTRODUCTORY LEARNING SCENARIO (HYPOTHETICAL HOOK)

**"You score 95/100. But can you still explain what you learned next week?"**

*Hypothetical learning scenario reflecting common educational experience: high exam marks often correlate with cramming rather than durable mastery.*

Grades are widely used across higher education to evaluate student performance, benchmark institutional progress, and determine academic opportunities. However, successful examination performance and meaningful learning are not necessarily the same thing.

**SSE-2601 RESEARCH THESIS:**

**"Grades are useful indicators of academic performance, but they cannot independently demonstrate the full depth of student learning."**

#### John Biggs (1996)

Constructive Alignment

Intended Learning Outcomes (ILOs), teaching-learning activities (TLAs), and assessment tasks must form an integrated, coherent system.

THEORETICAL FRAMEWORK MODEL

INTENDED LEARNING OUTCOMES → TEACHING & LEARNING ACTIVITIES → ASSESSMENT METHODS & RUBRICS

CRITICAL QUESTION:

***"Are we assessing what we actually intended to teach?"***

**Critical Balance:** Courses targeting critical reasoning must assess reasoning directly, not fall back on memorization tests.

CONSTRUCTIVE ALIGNMENT ARCHITECTURE (BIGGS, 1996) — MUTUALLY REINFORCED SYSTEM

| INTENDED OUTCOMES | TEACHING ACTIVITIES | ASSESSMENT METHODS |
|---|---|---|
| Higher-order reasoning & deep application goals | Active problem-posing, debate & practical labs | Authentic defense, systems design & rubrics |

*"Are we assessing what we actually teach? Assessment must directly measure intended cognitive depth."*

#### Paulo Freire (2000)

Problem-Posing vs. Banking Education

Critiques the 'banking model' where students act as passive containers of deposited facts. Champions problem-posing dialogue and active co-construction.

THEORETICAL FRAMEWORK MODEL

BANKING MODEL: Passive Fact Storing → VERSUS → PROBLEM-POSING: Active Critical Dialogue

CRITICAL QUESTION:

***"Are students passive recipients or active co-creators?"***

**Critical Balance:** Supports balance: active participation works best when grounded in rigorous teacher guidance and structure.

#### FORMATIVE FEEDBACK AS AN ENGINE FOR LEARNING

Assessment as Learning

Assessment must serve as a continuous learning instrument rather than a retroactive post-mortem grade.

| ASSESS | FEEDBACK | REFLECT | IMPROVE |
|---|---|---|---|
| Diagnostic drafts, peer reviews, low-stakes tests | Timely, specific, criteria-aligned guidance | Student self-interrogation & gap identification | Concrete revision before summative judgment |

*"Feedback is only valuable when students are afforded the structured opportunity to interpret and apply it."*

### SECTION 2 • THE CENTRAL PEDAGOGICAL METAPHOR

### THE FIVE DIMENSIONS OF COGNITIVE DEPTH

~10% Visible Evidence | ~90% Submerged Learning

SURFACE CRITERIA • WHAT WE CAN SEE

LEVEL 01 • VISIBLE — SURFACE • 0 m

**ACADEMIC PERFORMANCE**

- Examination scores & test points
- Correct formulaic answers
- Reproduced textbook definitions
- Completed assessment requirements

*The visible grade provides quantifiable evidence of performance in a specific, timed assessment task. However, it does not automatically reveal how deeply or flexibly the student understands underlying principles.*

Essential for baseline benchmarking, but only represents the tip of cognitive capability.

LAYER 01 • MEANING-MAKING

LEVEL 02 • SUBMERGED — DEPTH • -150 m

**CONCEPTUAL UNDERSTANDING**

- Explain ideas in your own words
- Connect related & disparate concepts
- Understand why mechanisms function
- Recognize principles across domains

*Meaningful learning transcends rote recitation. Students grasp structural relationships and can explain the foundational rationale without relying on memorized scripts.*

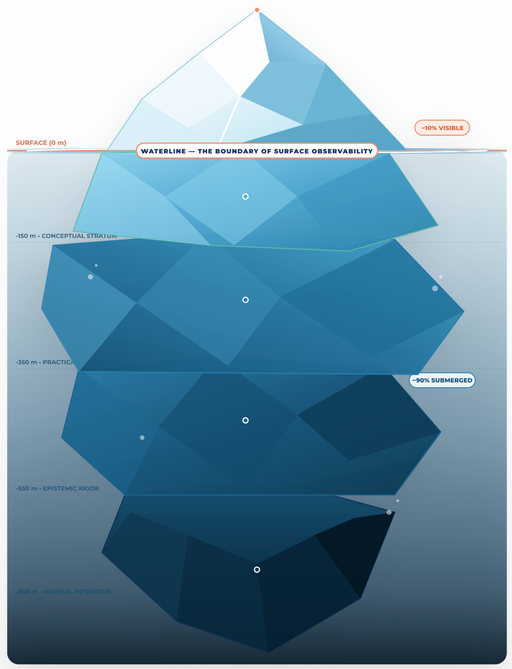
LAYER 03 • EPISTEMIC RIGOR

LEVEL 04 • SUBMERGED — DEPTH • -550 m

**CRITICAL THINKING**

- Question foundational assumptions
- Critically evaluate alternatives
- Defend methodological decisions
- Synthesize independent viewpoints

*Students actively interrogate ideas, critique methodologies, and adapt to ambiguity instead of treating knowledge as a static repository of unquestioned truths.*




LAYER 02 • ACTIVE DEPLOYMENT

LEVEL 03 • SUBMERGED — DEPTH • -350 m

**PRACTICAL APPLICATION**

- Apply theory to unfamiliar problems
- Solve realistic engineering tasks
- Justify chosen design trade-offs
- Diagnose system errors & anomalies

*Mastery becomes visible when students deploy theoretical knowledge to build, analyse, and troubleshoot real systems. Illustrated in Computer Architecture: understanding CPU ↔ RAM bus interactions rather than just defining registers.*

Contextual problem-solving validates genuine competence.

LAYER 04 • DURABLE RETENTION

LEVEL 05 • SUBMERGED — DEPTH • -800 m

**LONG-TERM RETENTION**

- Durable recall long after exam week
- Explain concepts spontaneously without notes
- Anchor new knowledge to existing schema
- Transfer insights into professional practice

*The true measure of learning is durability. Meaningful knowledge integrates into the student's cognitive framework, remaining accessible and actionable across semesters and careers.*

**Conceptual Principle:**
These five dimensions represent complementary aspects of cognitive depth, not a rigid sequential hierarchy. Grades reveal localized performance; authentic learning requires conceptual, practical, and epistemological roots.

**A HIGH GRADE IS VISIBLE. MEANINGFUL LEARNING GOES DEEPER.**

*"A high grade is visible evidence. True pedagogical mastery goes deeper."*

### CRITICAL COMPARISON: EXAMINATIONS VS. PROJECTS

Showing Both Sides

| TRADITIONAL EXAMINATIONS — STANDARDIZED | PRACTICAL & AUTHENTIC PROJECTS — AUTHENTIC |
|---|---|
| **WHAT THEY ASSESS WELL:** | **WHAT THEY REVEAL:** |
| • Efficient verification of foundational terminology & core facts<br>• Evaluates individual performance without collaborative obfuscation | • Direct demonstration of theoretical knowledge applied to real tools<br>• Fosters architectural decision-making and troubleshooting |
| **WHAT THEY MAY OVERLOOK:** | **CRITICAL LIMITATIONS:** |
| • Prone to temporary cram-and-forget memorization strategies<br>• Limited capacity to evaluate complex, open-ended system design | • Project completion does not automatically prove individual mastery<br>• Risk of 'free-riding' or copying boilerplate solutions from internet |

***Balanced Conclusion:*** *"Neither examination scores nor completed projects provide a complete picture of learning when considered in isolation. True rigor emerges from their deliberate combination."*

#### Grabinger & Dunlap (1995)

Rich Environments for Active Learning (REAL)

Promotes learning through authentic contexts, collaborative responsibility, dynamic problem-solving, and reflective inquiry.

THEORETICAL FRAMEWORK MODEL

AUTHENTIC CONTEXTS → COLLABORATIVE PROBLEM-SOLVING → STUDENT RESPONSIBILITY & REFLECTION

CRITICAL QUESTION:

***"Can students utilize knowledge outside the textbook sandbox?"***

**Critical Balance:** A completed project alone does not guarantee individual mastery; students must verbally defend their rationale.

### SECTION 3 • APPLICATION TO HIGHER EDUCATION

### FROM GRADES TO MEANINGFUL LEARNING

Balanced Ecosystem

FIVE COMPLEMENTARY ASSESSMENT MODALITIES

| Examinations | Practical Projects | Case Studies | Presentations & Q&A | Formative Feedback |
|---|---|---|---|---|
| DIAGNOSTIC & FOUNDATIONAL | APPLICATION & CREATION | ANALYTICAL REASONING | COMMUNICATION & DEFENSE | ITERATIVE IMPROVEMENT |
| Verify core theoretical knowledge, individual mastery, and fluency in basic terminology under controlled conditions. | Deploy principles in realistic environments, evaluate system trade-offs, and produce authentic artifacts. | Analyze complex, ill-structured dilemmas, balance competing engineering priorities, and justify solutions. | Articulate concepts verbally, respond to critical scrutiny, and defend technical decisions in real time. | Provide timely diagnostic feed-forward before high-stakes milestones, turning assessment into a learning tool. |

**COMPUTING SYSTEMS APPLICATION** — Computer Organisation and Architecture (SSE-2601 Case)

CPU (PROCESSING) — Registers & ALU Control

Bus Interaction — *Cache Hierarchy & Latency*

RAM (MEMORY) — Address Bus & Storage

**Rote Reproduction:** Memorizing textbook hardware definitions without understanding system dynamics.

**Deep Application:** Tracing bus latency bottlenecks and justifying architectural memory trade-offs.

EVIDENCE-INFORMED HE RECOMMENDATIONS

| 01 Constructive Alignment | 02 Dual Assessment Modality | 03 Active Dialogue & Defense | 04 Actionable Formative Loops |
|---|---|---|---|
| Align examination prompts and practical rubrics directly with intended higher-order learning outcomes. | Combine timed individual verification with authentic practical tasks and oral defenses. | Integrate peer feedback, classroom debate, and verbal justification into technical coursework. | Ensure draft milestones receive timely, actionable feedback before final summative grading. |

Measure not only what students can reproduce, but also what they can explain, apply, and retain.

### SECTION 4 • FINAL CONCLUSION & PEDAGOGICAL POSITION

SSE-2601 Consensus

A high grade may indicate that a student has mastered the material, but it is not sufficient proof of deep understanding. Its meaning depends on the assessment task, intended learning outcomes, and knowledge demonstrated.

Meaningful learning becomes evident when students can explain ideas, apply theory in unfamiliar situations, critically evaluate solutions, and retain knowledge across time. Rather than abandoning examinations, higher education must synthesize multiple assessment modalities, active dialogue, and actionable formative feedback.

**"Grades should measure learning — not define its limits."**

A HIGH GRADE IS VISIBLE. MEANINGFUL LEARNING GOES DEEPER.

### SECTION 5 • REFERENCES (APA 7TH EDITION)

Verified Scholarly Sources

**Biggs, J.** (1996). *Enhancing teaching through constructive alignment.* Higher Education, 32(3), 347–364. https://doi.org/10.1007/BF00138871

**Freire, P.** (2000). *Pedagogy of the oppressed (30th anniversary ed.).* Continuum.

**Grabinger, R. S., & Dunlap, J. C.** (1995). *Rich environments for active learning: A definition.* Research in Learning Technology, 3(2), 5–34. https://doi.org/10.1080/0968776950030202

**Magnússon, G., & Rytzler, J.** (2022). *Towards a pedagogy of higher education: The Bologna process, Didaktik and teaching.* Routledge. https://doi.org/10.4324/9781003054160

INDIVIDUAL RESEARCH CONTRIBUTIONS (SSE-2601)

**Nikita Shulmin**
Academic performance, memorisation-based assessment, practical assessment, and feedback.

**Dana Kenzhebayeva**
Constructive alignment, practical judgment, educational philosophy, and the balance between structure and flexibility.

**Islam Akhmetov**
Practical learning, formative feedback, and theoretical knowledge applied to Computer Organisation and Architecture.

**Abzhappar Abylaikhan**
Interactive teaching, student participation, classroom discussions, and group-based mini-lessons.